

3.3 Unit 3 Investigative and Practical Skills in AS Chemistry

Candidates should carry out experimental and investigative activities in order to develop their practical skills. Experimental and investigative activities should be set in contexts appropriate to, and reflect the demand of, the AS content. These activities should allow candidates to use their knowledge and understanding of Chemistry in planning, carrying out, analysing and evaluating their work.

The content of Units 1 and 2 provides a basis for different practical topics which may be used for experimental and investigative skills. The experience of dealing with such activities will develop the skills required for the assessment of these skills in the Unit. Examples of suitable experiments that could be considered throughout the course will be provided in the Teacher Resource Bank.

It is expected that candidates will be able to use and be familiar with 'standard' laboratory equipment which is deemed suitable at AS level, throughout their experiences of carrying out their practical activities.

The skills developed in the course of their practical activities are elaborated further in the *How Science Works* section of this specification (see Section 3.7).

In the course of their experimental work, candidates should learn to

- demonstrate and describe ethical, safe and skilful practical techniques
- process and select appropriate qualitative and quantitative methods
- make, record and communicate reliable and valid observations
- make measurements with appropriate precision and accuracy
- analyse, interpret, explain and evaluate the methodology, results and impact of their own and others' experimental and investigative activities in a variety of ways.

The Centre-Assessed Option, Route T

The practical and investigative skills will be centre assessed through

- Investigative Skills Assignment (ISA)
- Practical Skills Assessment (PSA).

The ISA will require candidates to undertake practical work, collect and process data and use it to answer questions in a written test (ISA test) (see Section 3.8).

The PSA requires centre assessment, throughout the AS course, of the candidate's ability to follow and undertake certain standard practical activities across the three areas of Chemistry: Inorganic, Organic and Physical. These practical activities are listed on the next page.

The Externally Marked Option, Route X

The practical and investigative skills will be externally assessed by AQA through

- Practical Skills Verification (PSV) and
- Externally Marked Practical Assignment (EMPA)

The PSV will require candidates to undertake the practical activities identified on the next page, in order to gain experience of a wide range of practical skills. These activities must allow a candidate suitable opportunity to demonstrate safe and skilful practice, as well as producing reliable and valid observations. AQA will require teacher verification, by means of a tick box on the Candidate Record Form, that a candidate is experienced in these skills.

The EMPA, in a similar way to the ISA, will require candidates to undertake practical work, collect and process data and use it to answer questions in a written test (EMPA test) (see Section 3.8).

Practical Skills Assessment

The PSA is designed to credit candidates for the practical work they undertake naturally as part of the course. Further information is provided in Section 3.8, but candidates should complete at least two practical activities from each of the three areas of Chemistry listed below.

AS Inorganic Chemistry

Task	Possible context
Make up a volumetric solution	The preparation of a standard solution of sodium carbonate
Carry out a simple acid-base titration	Determine the concentration of unknown hydrochloric acid by titration
Carry out some inorganic tests	Tests for anions

AS Physical Chemistry

Task	Possible context
Measure an enthalpy change	Use Hess's law to find an unknown enthalpy change, such as the reaction of anhydrous copper(II) sulfate with water to produce hydrated crystals
Determine the M_r of a volatile liquid or the M_r of a gas	Determine the M_r of hexane or the M_r of carbon dioxide
Investigate how the rate of a reaction changes with temperature.	Investigate the rate of reaction of sodium thiosulfate with acid at different temperatures

AS Organic Chemistry

Task	Possible context
Distil a product from a reaction	The preparation of ethanal from the oxidation of ethanol or the preparation of cyclohexene from the dehydration of cyclohexanol
Carry out some organic tests	Tests for alkene, alcohol, acid, aldehyde
Investigate the combustion of alcohols	Use a calorimetric method to measure the enthalpies of combustion in an homologous series of alcohols